

Predictive Modelling for a Telecom Marketing Campaign

Statistical Thinking for Data Science — Assessment Task 3

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EXECUTIVE SUMMARY

This report develops predictive models for a telecom subscription campaign and identifies the customer segments most likely to convert. Three classifiers (Logistic Regression, Random Forest, Gradient Boosting) were trained under two feature setups: a realistic pre-contact one used for outbound targeting, and a full one used only as a benchmark since it includes call duration. The main result is that calling only the top 10% of customers ranked by the model would catch 47% of all future subscribers at a 53% conversion rate — a $4.7\times$ lift over the 11.3% baseline. The analysis points to four clear strategies for the next campaign: target customers during low interest rate periods, focus on March, October and December, re-contact previous responders, and prefer the cellular channel over the fixed line.

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1.INTRODUCTION AND PROBLEM FORMULATION

Marketing campaigns in telecom are expensive when poorly targeted, and most contacted customers end up not subscribing. Using predictive models to focus on the customers most likely to convert has become a common way to deal with this (Ngai et al., 2009). A preliminary exploratory analysis of 41,180 customer interactions revealed a strong class imbalance (only 11.3% of customers subscribed) and showed that the variable *duration* is only known after the call, which makes it unusable for any realistic pre-contact prediction (Moro et al., 2014). This report builds on those findings to develop predictive models that can answer two business questions: which customers should be prioritised, and what strategies follow from the model results.

The analysis addresses three questions:

- Q1.** Which customer and contextual features predict subscription?
- Q2.** How much predictive power is lost when post-contact information is excluded?
- Q3.** What concrete strategies can be derived from the parametric estimates?

The audience is twofold: the marketing operations team (who needs to know whom to call) and senior decision-makers (who need strategic recommendations).

2. METHODOLOGY

2.1 Feature configuration

The Logistic Regression assumes that predictors are not perfectly collinear, since highly correlated features make the coefficients unstable and hard to interpret. A look at the correlation matrix of the macroeconomic indicators showed pairwise correlations above 0.91 between *euribor3m*, *emp.var.rate* and *nr.employed*. To handle this, only *euribor3m* was kept among the three. A Variance Inflation Factor (VIF) check on the final set of numeric predictors confirmed that this choice was enough: all remaining VIFs sit well below 3 (*euribor3m* = 2.69, *cons.price.idx* = 2.04, the others below 1.5), comfortably under the usual cut-off of 10 that flags severe multicollinearity. The variable *pdays* was also dropped because it was mostly missing and the engineered *previously_contacted* already captures the same information more cleanly.

Two prediction setups were then defined to address the central methodological question of this analysis: can a useful targeting model be built using only the information available before a customer is called? Config A ("pre-contact") uses the 17 features that are known in advance — demographics, prior campaign history, contact channel and macroeconomic context. Config B ("full") adds *duration*, the length of the call. While *duration* is by far the strongest single correlate of subscription, it is only observable during the call and therefore cannot be used to decide whom to contact. Comparing the two configurations quantifies the predictive cost of removing this leaked signal and confirms whether the pre-contact features alone carry enough information for outbound targeting.

2.2 Model selection

Three models with increasing complexity were selected to compare both predictive performance and interpretability — a key concern when the goal is to derive actionable strategies, not just predictions.

Logistic Regression (parametric) is the interpretable baseline. Its coefficients translate directly into odds ratios, which makes it easy to communicate the effect of each feature to a non-technical audience. As a linear model, it may miss interactions, but this simplicity is an advantage when the model has to be explained or audited.

Random Forest (non-parametric, bagging) was added to capture non-linear effects and feature interactions that the Logistic Regression cannot represent. By averaging many decision trees trained on bootstrap samples, it reduces variance and stays robust to outliers and irrelevant features.

Gradient Boosting (non-parametric, boosting) is widely considered state-of-the-art on tabular data. Unlike Random Forest, trees are built sequentially, each one correcting the errors of the previous ensemble — usually a stronger fit, but at the cost of interpretability.

Hyperparameters were tuned through a 5-fold cross-validated grid search, with PR-AUC as the scoring metric. For the Logistic Regression, the regularisation strength C was tuned and the grid selected a strong regularisation ($C = 0.01$) — consistent with the high-dimensional space created by one-hot encoding, where smaller coefficients help avoid fitting noise. For the Random Forest, the grid covered the number of trees, maximum depth, and minimum samples per split. For the Gradient Boosting, the learning rate, maximum depth, and number of iterations were tuned together. The final values are reported in the code appendix. Tuning was only done on the pre-contact configuration, since the full version serves as a benchmark and is not meant to be deployed.

2.3 Handling class imbalance and model estimation

The dataset is strongly imbalanced, with only 11.3% positive cases. Without correction, a model can reach a misleadingly high accuracy just by predicting "no" for everyone while completely missing the subscribers. To make this concrete, two dummy baselines were trained alongside the main models: a "most frequent" classifier that always predicts "no", and a "stratified" classifier that predicts proportionally to the observed frequencies. The most-frequent dummy reaches 88.7% accuracy but identifies zero subscribers — a hard reference point that any useful model must clearly beat on recall, F1, or PR-AUC.

In the actual models, *class_weight='balanced'* was applied across all three. It re-weights the loss inversely to class frequencies, so the optimiser gives equal importance to both classes instead of being dominated by the majority. Numeric features were standardised (necessary for the Logistic Regression, harmless for the tree-based models), and categorical features one-hot encoded with the first category dropped to avoid collinearity.

The Logistic Regression itself is fitted by Maximum Likelihood Estimation (MLE) with L2 (Ridge) regularisation. MLE finds the coefficients that best fit the observed outcomes, and the L2 penalty shrinks them toward zero to reduce variance and limit overfitting. This matters here because one-hot encoding of the ten categorical variables produces around 50 dummy columns, so the risk of fitting noise is real.

3. Results and Comparative Analysis

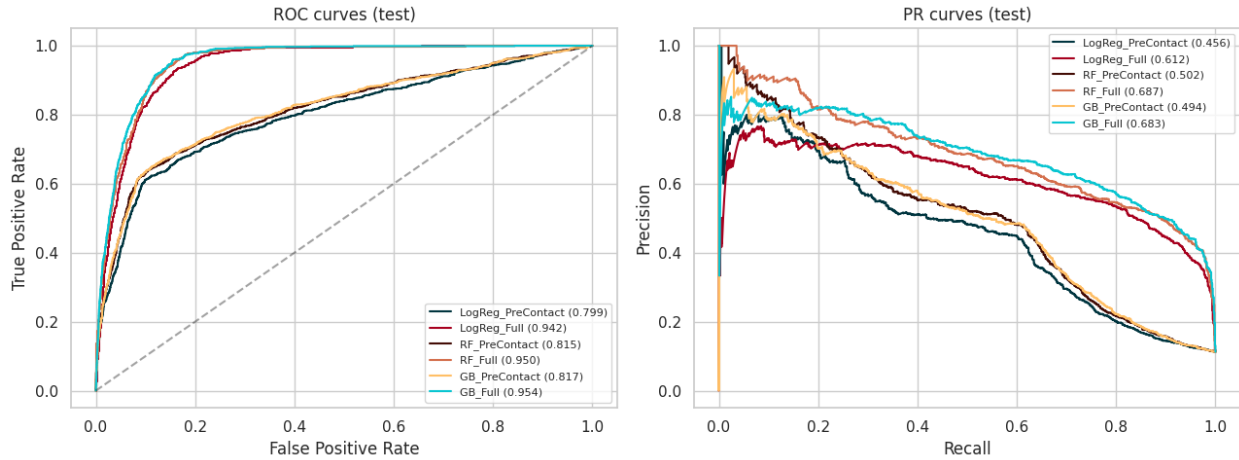
Table 1 shows the test set performance, including the two dummy baselines. The "most frequent" classifier reaches 89% accuracy without identifying a single subscriber — a good reminder of why accuracy alone is not a reliable metric on this dataset. All three trained models clearly outperform both dummies on every metric that accounts for the positive class, so they are picking up real signal and not just majority-class behaviour.

Table 1 — Test set performance (n = 8,234)

Model	ROC-AUC	PR-AUC	F1	Recall	Precision
Dummy (most frequent)	0.50	0.11	0.00	0.00	0.00
Dummy (stratified)	0.50	0.11	0.12	0.12	0.12
LogReg — Pre-contact	0.80	0.46	0.45	0.66	0.34
RF — Pre-contact	0.81	0.50	0.52	0.64	0.44
GBM — Pre-contact	0.82	0.49	0.36	0.25	0.69
LogReg — Full	0.94	0.61	0.60	0.90	0.45
RF — Full	0.95	0.69	0.63	0.91	0.48
GBM — Full	0.95	0.68	0.61	0.56	0.67

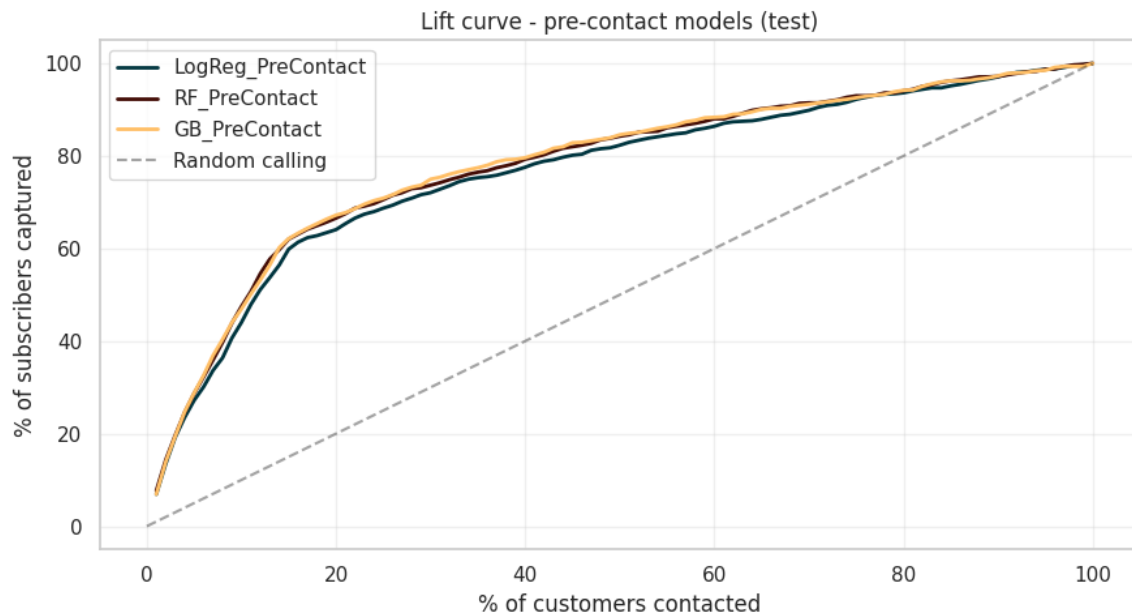
Two things stand out. First, the three pre-contact models perform very closely on PR-AUC (between 0.46 and 0.50), with the Random Forest slightly ahead. The tuned GBM ends up at the highest precision (0.69) but only catches 25% of the actual subscribers — it predicts fewer "yes" than the other models but is more often right when it does. This is mostly an effect of how the default threshold (0.5) interacts with the tuned model, not a sign that one model is fundamentally worse. PR-AUC, which does not depend on the threshold, stays close across the three. The takeaway is that the simpler Logistic Regression is nearly as good as the more complex models, which is a strong argument for using it in production since it is much easier to explain. Second, the gap between the pre-contact and full configurations is large (+0.13 ROC-AUC, +0.20 PR-AUC). This confirms that *duration* carries a lot of information, but also that it cannot be used in practice since it is only known after the call. [Figure 1: ROC and PR curves]

Figure 1. ROC curves (left) and Precision-Recall curves (right)



To make the performance more concrete for the business, Figure 2 shows the lift curve. Customers are ranked by predicted probability and contacted from the highest down. Contacting just the top 10% would catch 47% of subscribers at a **53%** conversion rate (a 4.7× lift over the baseline), and the top 20% would catch 67% at 38%. The curve flattens clearly after 20%, which suggests a good cut-off for budget allocation. [Figure 2: Lift curve]

Figure 2. Lift curve on the test set — % of subscribers captured against % of customers contacted, ordered by predicted probability.



4. Interpretation and Business Insights

4.1 Parametric estimates

Table 2 shows the strongest coefficients from the Logistic Regression. Odds ratios above 1 increase the probability of subscribing; below 1 they decrease it.

Table 2 — Top Logistic Regression coefficients (pre-contact)

Feature	Coefficient	Odds Ratio
previously_contacted	+0.78	2.17
poutcome = success	+0.67	1.95
month = March	+0.64	1.89
month = October	+0.35	1.42
poutcome = nonexistent	+0.29	1.34
job = retired	+0.27	1.31
cons.price.idx	+0.24	1.27
job = student	+0.18	1.20
euribor3m	−0.90	0.41
month = May	−0.67	0.51
contact = telephone	−0.36	0.70
default = unknown	−0.21	0.81

The Random Forest and GBM feature importances pick out the same top variables (*euribor3m*, *cons.price.idx*, *previously_contacted*, *month*, *contact*), so the three models tell a consistent story.

4.2 Four business insights

Insight 1 — Macroeconomic timing matters more than customer profile. *Euribor3m* is the strongest predictor in all three models — largest coefficient in the Logistic Regression (-0.90 , odds ratio 0.41), and clear first place in both the Random Forest (importance 0.26) and the GBM (0.25), far above any demographic variable. A one-standard-deviation drop in the rate multiplies the odds of subscribing by about $2.4\times$, well above anything tied to job, age or education. When rates are low, customers have fewer attractive alternatives and commit more easily. The real targeting question is therefore *when* to call, not *who*. Focusing campaigns on favourable rate periods could roughly halve the cost per acquisition without changing the customer pool.

Insight 2 — Re-contacting previous responders is the easiest lever to pull. Two contact-history variables stand out: *previously_contacted* (odds ratio 2.17) and *poutcome = success* (1.95), both also highly ranked by the Random Forest and GBM. Customers who already engaged with the company — especially those who said yes to an earlier campaign — are the highest-yield segment the models can flag before any new call. A simple re-engagement workflow (contacting them first, or routing them to more experienced agents) would give an immediate uplift at almost no cost, and it is the simplest of the four recommendations to act on.

Insight 3 — The month of contact matters, but with caveats. March has one of the highest odds ratios in the model (1.89), followed by October (1.42), while May has the strongest negative effect (0.51). These numbers should be read carefully: since interest rates and macroeconomic conditions shift over time, the month effect likely captures part of the context discussed in Insight 1 rather than a purely seasonal pattern. Still, the signal is consistent across the three models, so it is worth using when planning the calendar — shifting budget from weaker to stronger months could improve conversion without changing the targeting itself.

Insight 4 — Cellular clearly beats fixed-line. *Contact = telephone* has a coefficient of -0.36 , a 30% drop in odds compared to cellular, and the effect is consistent across all three models. Two explanations are plausible: behavioural (mobile users skew younger and more responsive) or operational (fixed-line calls likely have lower answer rates and shorter conversations). Either way, the recommendation is the same — shifting more outbound capacity toward cellular is operationally trivial, doesn't need a model in production, and should give a visible uplift on the next campaign.

5. Limitations and Conclusion

A few things should be kept in mind. The class imbalance is only partly handled by class weighting: precision in pre-contact mode stays around 0.40, so about one in 2.5 contacted customers will actually subscribe — better than the one-in-nine baseline, but far from perfect. The macroeconomic features also come from a specific period, so the strong effect of *euribor3m* may not hold in a very different rate environment. And the whole analysis is correlational: any causal claim, like switching everyone to cellular, would need an actual experiment.

On the modelling side, the grid search was kept fairly narrow to stay within memory limits, and the best C for the Logistic Regression landed at the lower bound of the grid (0.01) — so even stronger regularisation could be worth testing. The decision threshold was also left at 0.5, which makes the tuned GBM look weaker on F1 and recall than it really is. A calibrated threshold or a nested cross-validation would give a more precise view.

A few additional data sources would also help. Following the same customers across several campaigns would help separate the macroeconomic effect from any real seasonal pattern in Insight 3. A randomised test on the cellular versus fixed-line comparison would be needed before claiming a causal effect. And knowing more about what was said during past calls would help check whether the *previously_contacted* effect comes from genuine interest or just from the company picking the easier leads.

Within these limits, the analysis still gives a clear and usable direction. The pre-contact Logistic Regression is nearly as accurate as the Gradient Boosting model and stays easy to interpret, which makes it a good fit for an operational targeting tool. Combined with the four insights, it gives both a way to rank customers and a set of concrete levers — timing the campaign with macroeconomic conditions, prioritising re-engagement, adjusting the calendar, and shifting toward cellular — that can be put to work in the next campaign.

REFERENCES

Moro, S., Cortez, P., & Rita, P. (2014). A data-driven approach to predict the success of bank telemarketing. *Decision Support Systems*, 62, 22–31.

<https://doi.org/10.1016/j.dss.2014.03.001>

Ngai, E. W. T., Xiu, L., & Chau, D. C. K. (2009). Application of data mining techniques in customer relationship management: A literature review and classification. *Expert Systems with Applications*, 36(2),

2592–2602. <https://doi.org/10.1016/j.eswa.2008.02.021>

APPENDIX

Appendix 1- Generative AI Usage Acknowledgement This report was prepared in accordance with UTS guidelines on the use of generative AI tools. AI assistance was used for APA reference formatting, wording suggestions, and consistency review between the literature review, methods and results. All content — including the selection of cited papers, the design of the ablation study, the interpretation of results and the final wording — was produced, reviewed and validated by the authors.